Pitch Script and Q&A

A five-minute script, a demo plan and answers to likely judge questions

CoolRoute: Pedestrian Routing with Shade and Heat-Exposure Budgets

1. Five-minute pitch script

Time	Say	Show
0:00 to 0:40	**Problem.** 'The shortest walk is often the hottest. In summer, a few minutes in the sun or standing at an open crossing makes a big difference, and for people who cannot use steps the choice is even smaller. Map apps count distance, not heat.'	Slide: one hot, bare street
0:40 to 1:20	**Idea.** 'CoolRoute gives two routes: the shortest, and the shortest that fits your heat budget. Then it tells you, in one sentence, what the cooler route costs and gives.'	Prototype: both routes at 1 pm
1:20 to 2:30	**Live demo.** Follow the demo steps in section 2.	Prototype
2:30 to 3:30	**How it works.** 'Streets are a graph. Heat is length x (1 - shade) x sun. We give heat a price, lambda, and search for the lambda that fits the budget. Step-free is never traded away.'	File 06: formula and trade-off curve
3:30 to 4:15	**Proof.** '25 automated checks. JavaScript and Python agree on 60 scenarios. On 148 random trips, the typical route is about 40% cooler for about 3% more distance. And we show you where it fails: a 60% budget is met on about half of trips.'	File 07: histogram and test table
4:15 to 5:00	**Future.** 'Next: real Jaipur data, shade model on real trees and buildings, a pilot on campus, city licences and a route API.'	File 08: revenue chart

2. Live demo: click by click (about 70 seconds)

#	Do	Say
1	Open the prototype on the 'Hot afternoon' preset (1 pm, 60% budget)	'Orange is the shortest route. Teal is CoolRoute.'
2	Point to the verdict sentence	'23 metres longer, about 40% less heat. It keeps to the green, tree-lined avenue.'
3	Drag the time slider to evening	'As the sun weakens, the heat gap shrinks. At night both routes are the same.'
4	Go back to 1 pm. Slide the budget down to 30%	'A stricter budget costs more distance. When it cannot be met, we say so, and show the coolest route.'
5	Turn on Step-free	'Step-free is a hard rule. Routes use only step-free segments, even if hotter.'
6	Choose 'Close footpath' and tap a street on the route	'A closed footpath is avoided straight away.'
7	Point to the dashed streets and the warning list	'Dashed means missing shade data. We show the gap and call the result approximate.'

3. Likely judge questions and answers

Q1. Why not just use Google Maps?

Map apps optimise distance and time. They do not count shade or time-of-day heat, and they do not let you set a heat budget. We add that, and we explain the trade-off in one sentence.

Q2. How do you measure heat?

Heat exposure is metres walked in full sun: length x (1 - shade) x sun strength, plus waiting at crossings. It is a relative score for comparing routes.

Q3. Is the data real?

The prototype uses a synthetic neighbourhood so the demo is reliable and we can test failure cases. The real-city pipeline uses OpenStreetMap, satellite tree cover and building heights, described in file 05.

Q4. Is the ML real?

Yes, it runs and is tested, but it is trained on synthetic data built from the shadow formula. It shows the pipeline and the confidence range. Real accuracy needs real data, and we say so.

Q5. How do you solve the heat budget?

We give heat a price lambda and minimise distance + lambda x heat. We search lambda by bisection until the route fits the budget. It is fast and explainable.

Q6. Is that optimal?

It finds a route on the efficient frontier and always fits the budget when such a route exists there. It may miss the exact optimum between frontier points. Label-setting is our planned exact upgrade.

Q7. What if the budget cannot be met?

We return the coolest route and show a warning. On our 148-trip test, a 60% budget was met on about half of trips, and the app tells the user when it is not.

Q8. What about accessibility?

Step-free is a hard rule. Those segments are removed before the search, so no route ever uses them. If no step-free route exists, we say so.

Q9. What if data is missing?

Each street piece has a confidence value. Low confidence is drawn dashed, counted in the warnings, and the heat value is called approximate.

Q10. Do you give health advice?

No. The score compares routes. We say on every result that it is not a medical risk.

Q11. What if the cooler route is very long?

A warning appears when it adds more than 20% distance, and the user can raise the heat budget.

Q12. How fast is it?

Milliseconds on our map. For a real city we limit the search area around A and B and use caching or contraction hierarchies.

Q13. How did you test it?

13 engine tests, 12 backend tests, a check that JavaScript and Python give identical results on 60 scenarios, and a 148-trip experiment. All in file 07.

Q14. How will you make money?

City licences, a route API for apps and a premium citizen app. Figures are assumptions to validate (file 08).

Q15. What would you do with more time?

Load Jaipur from OpenStreetMap, train the shade model on real tree and building data, walk the routes to validate, and add live weather from Open-Meteo.

4. Do and do not

Do	Do not
Say clearly that the data is synthetic and the ML demo is trained on synthetic data	Claim real-world accuracy
Show a failure case (budget not met, large detour) before a judge asks	Hide the cases where the cooler route is too long
Use the words 'comparison score'	Say 'health risk' or 'heat stroke risk'
Keep the browser prototype open as the backup	Depend on live internet or a sleeping free server
Tell the trade-off in one sentence	Read formulas aloud; show the chart instead

5. Checklist before you present

- Open the prototype and test the sliders on the presentation laptop.
- Keep a screenshot of the main view as a backup.
- Print or open files 06 and 07 to answer technical questions.
- Practise the script with a timer. Aim for 4:30 so you have a buffer.
- Agree who speaks about product, who about technology, who about business.